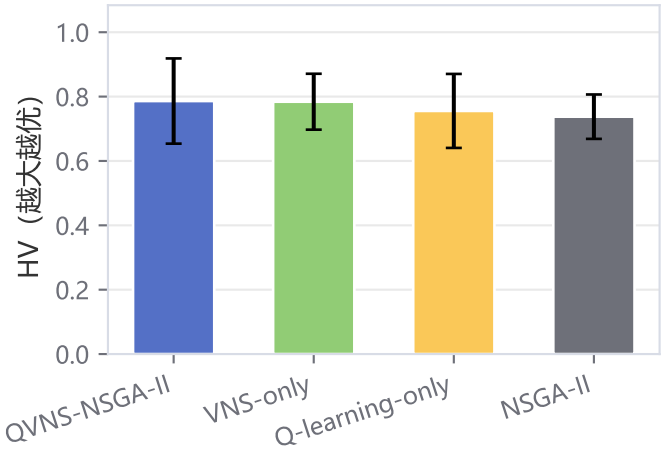
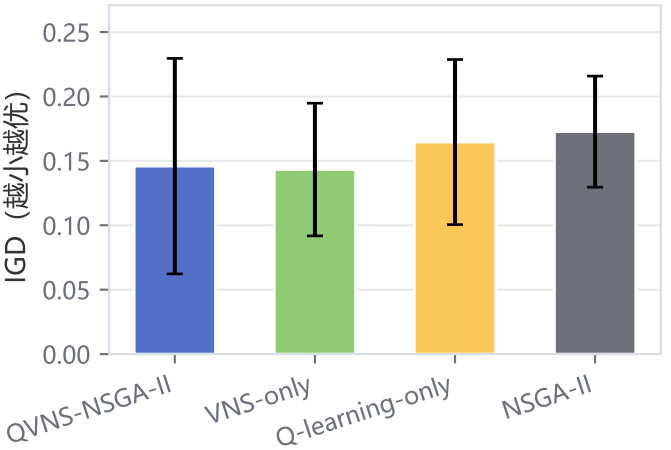
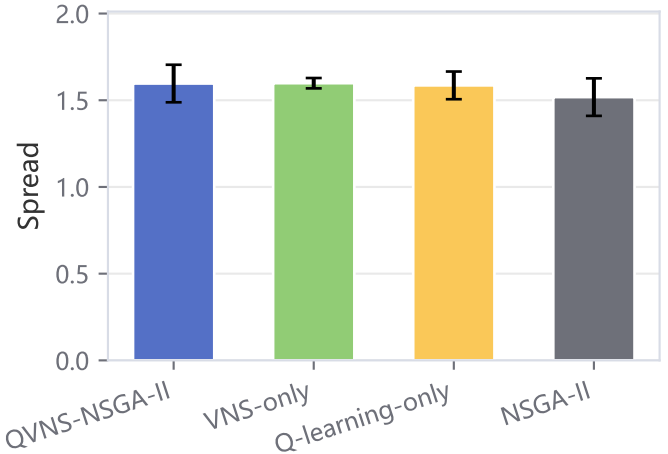
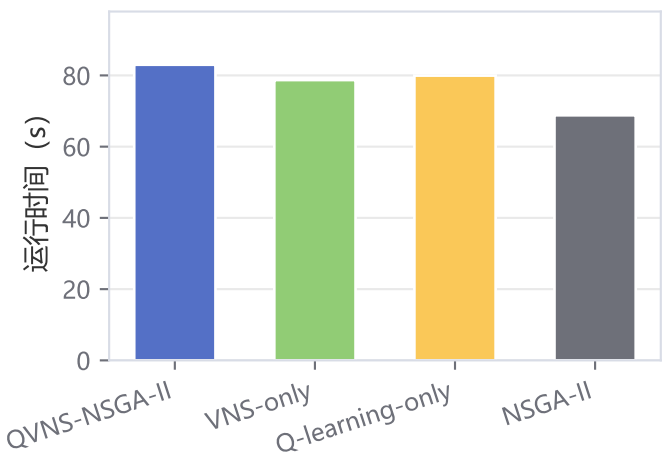

HV（越大越优）
1.0
0.8
0.6
0.4
0.2
0.0
QVNS-NSGA-II
VNS-only
Q-learning-only
NSGA-II


IGD（越小越优）
0.25
0.20
0.15
0.10
0.05
0.00
QVNS-NSGA-II
VNS-only
Q-learning-only
NSGA-II


Spread
2.0
1.5
1.0
0.5
0.0
QVNS-NSGA-II
VNS-only
Q-learning-only
NSGA-II


运行时间（s）
80
60
40
20
0
QVNS-NSGA-II
VNS-only
Q-learning-only
NSGA-II